EXP NO :-11
PROGRAM:-

```
clc;
clear;
close all;

% Experiment 1 filter
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

% PDF values
B = 8;
Xmax = 1;
A = 0.9;

% Normalized input
Nsample = 200;
x = A*ones(1,Nsample);

y = filter(b,a,x);

% Q1.7 quantization
S = 2^(B-1);
yq = round(y*S)/S;

overflow = abs(yq) > Xmax;

fprintf('Number of overflow samples = %d\n',sum(overflow));
fprintf('Peak output = %f\n',max(abs(yq)));
```

```
figure;
plot(yq);
hold on;
yline(1,'r--');
yline(-1,'r--');
grid on;
xlabel('Sample');
ylabel('Amplitude');
title('Overflow Response - Q1.7');

clc;
clear;
close all;

% Experiment 1 Butterworth LPF
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

B = 8;
S = 2^(B-1);

% PDF input amplitude
x = 0.9*ones(1,200);

y = filter(b,a,x);

% Fixed-point integer representation
yi = round(y*S);

% Saturation
ysat = min(max(yi,-128),127)/S;
```

```
% Wrap-around
ywrap = mod(yi+128,256)-128;
ywrap = ywrap/S;

figure;

subplot(2,1,1);
plot(ysat);
grid on;
title('Saturation Output');
ylabel('Amplitude');

subplot(2,1,2);
plot(ywrap);
grid on;
title('Wrap-Around Output');
xlabel('Sample');
ylabel('Amplitude');

clc;
clear;
close all;

% Experiment 1 filter
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

% PDF values
A = 0.9;
alpha = 0.5;
```

```
x = A*ones(1,200);

% Original
y = filter(b,a,x);

% Scaled
xs = alpha*x;
ys = filter(b,a,xs);

Xpeak = max(abs(y));
Xspeak = max(abs(ys));

H = 20*log10(1/Xpeak);
Hs = 20*log10(1/Xspeak);

fprintf('Original peak = %f\n',Xpeak);
fprintf('Scaled peak   = %f\n',Xspeak);

fprintf('Original headroom = %.4f dB\n',H);
fprintf('Scaled headroom   = %.4f dB\n',Hs);

figure;
plot(y);
hold on;
plot(ys,'r--');
grid on;
legend('Unscaled','Scaled');
xlabel('Sample');
ylabel('Amplitude');
title('Effect of Signal Scaling');

clc;
clear;
close all;

% Experiment 1 Butterworth LPF
```

```
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

% PDF values
A = 0.9;
alpha = 0.5;

x = A*ones(1,200);
xs = alpha*x;

y = filter(b,a,x);
ys = filter(b,a,xs);

% Q1.7
S = 128;

yq = round(y*S)/S;
ysq = round(ys*S)/S;

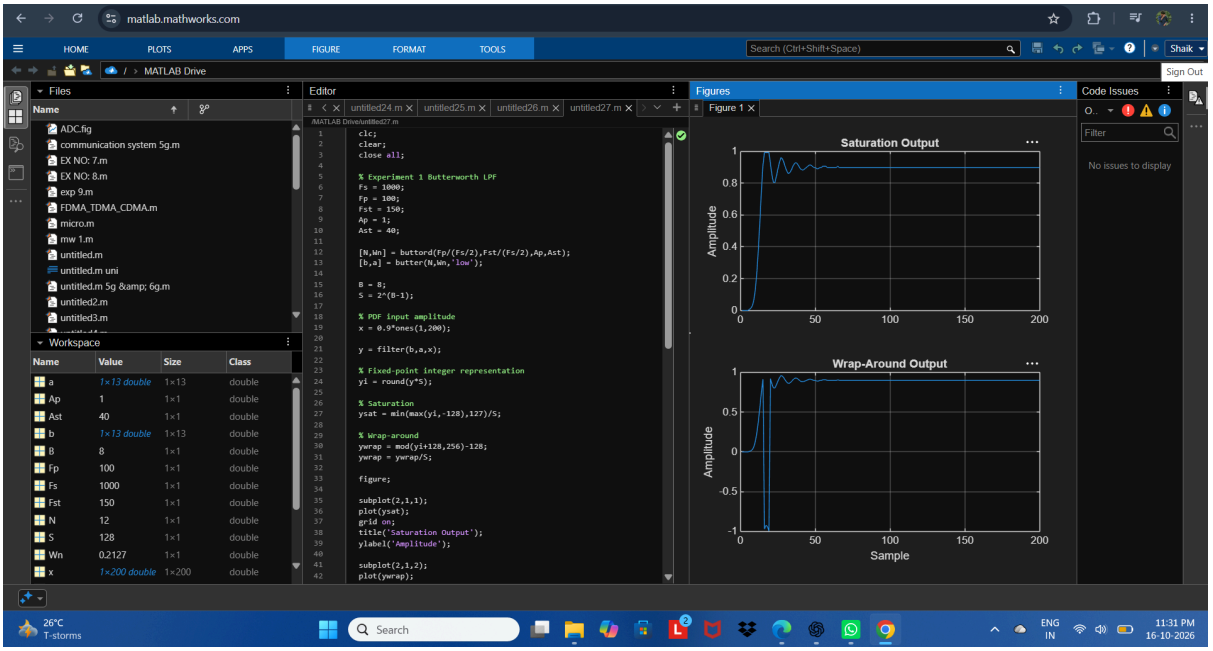
overflow_unscaled = sum(abs(yq)>1);
overflow_scaled = sum(abs(ysq)>1);

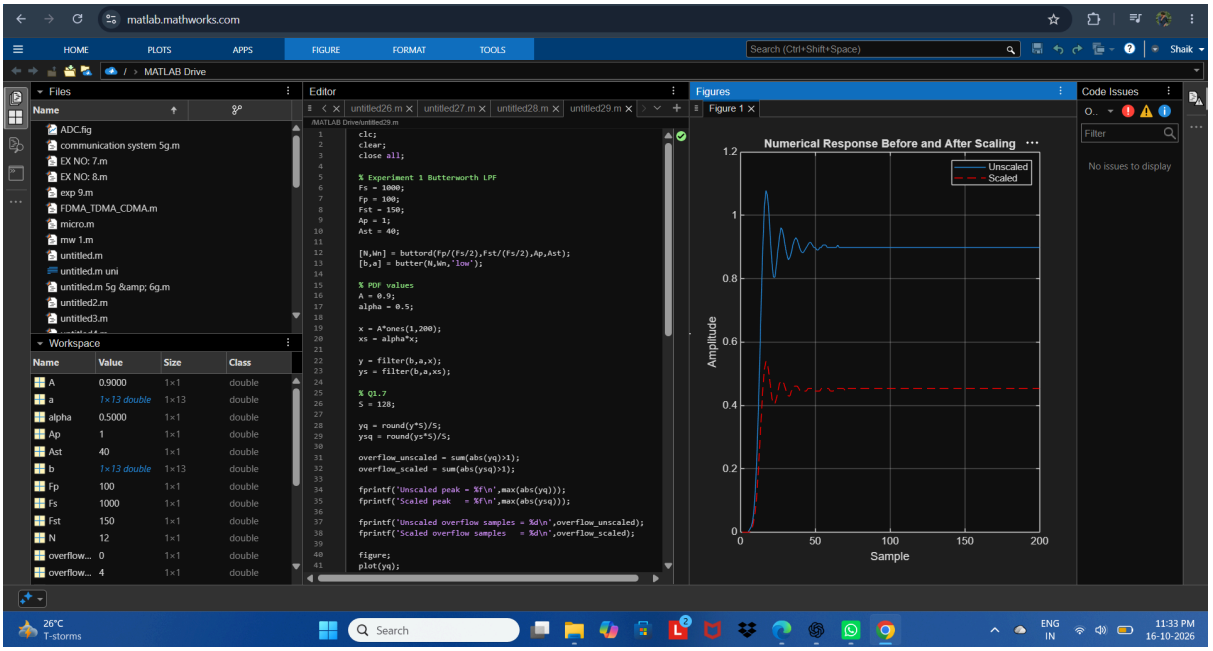
fprintf('Unscaled peak = %f\n',max(abs(yq)));
fprintf('Scaled peak   = %f\n',max(abs(ysq)));

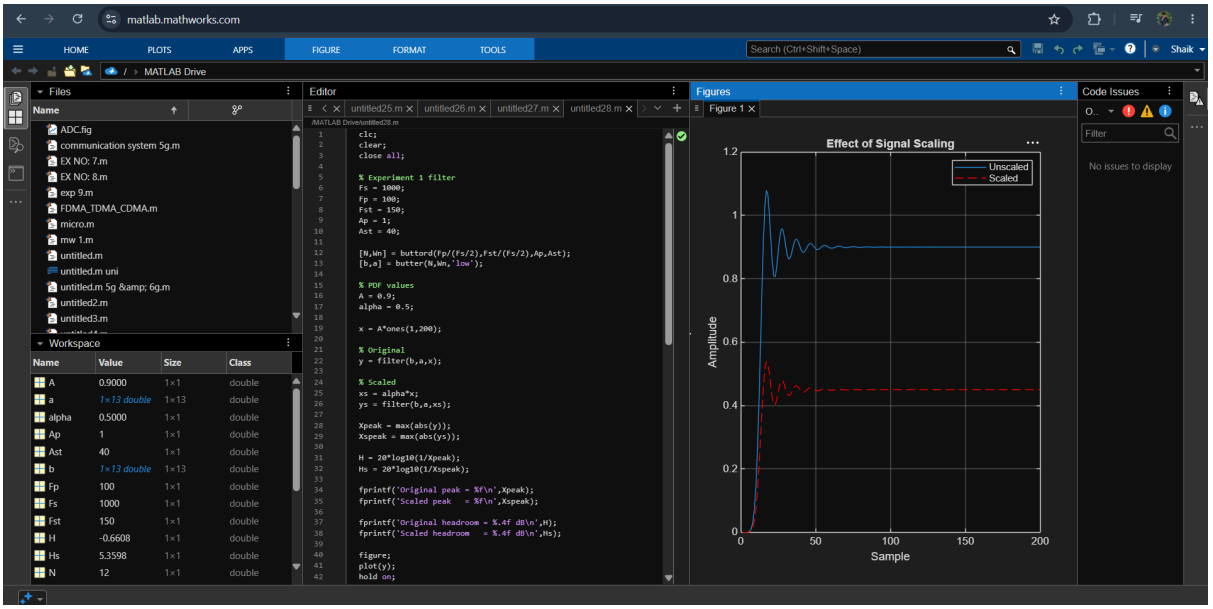
fprintf('Unscaled overflow samples = %d\n',overflow_unscaled);
fprintf('Scaled overflow samples   = %d\n',overflow_scaled);

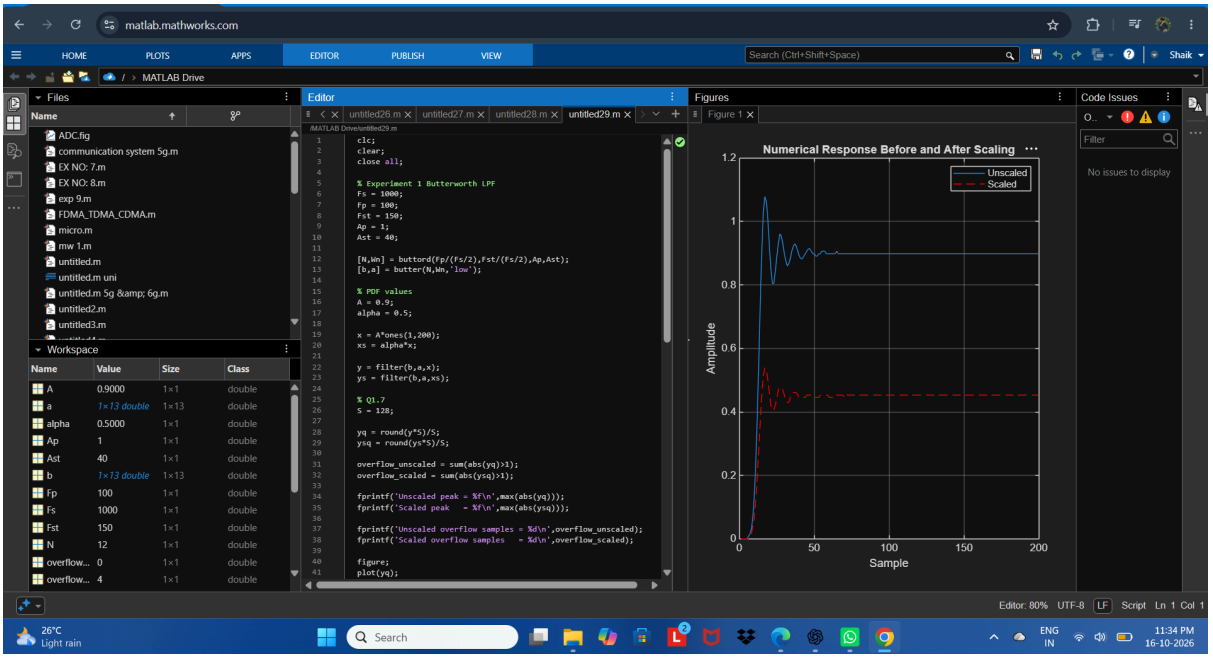
figure;
plot(yq);
hold on;
```

```
plot(ysq,'r--');
grid on;
legend('Unscaled','Scaled');
xlabel('Sample');
ylabel('Amplitude');
title('Numerical Response Before and After Scaling');
```






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EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
A 0.9000 1×1 double
a 1×13 double 1×13 double
alpha 0.5000 1×1 double
Ap 1 1×1 double
Ast 40 1×1 double
b 1×13 double 1×13 double
Fp 100 1×1 double
Fs 1000 1×1 double
Fst 150 1×1 double
H -0.6608 1×1 double
Hs 5.3598 1×1 double
N 12 1×1 double
Editor
untitled25.m
untitled26.m
untitled27.m
untitled28.m
/MATLAB Drive/untitled28.m
clc;
clear;
close all;
% Experiment 1 filter
Fs = 1000;
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Fst = 150;
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[b,a] = butter(N,Wn,'low');
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x = A*ones(1,200);
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% Scaled
xs = alpha*x;
ys = filter(b,a,xs);
Xpeak = max(abs(y));
Xspeak = max(abs(ys));
H = 20*log10(1/Xpeak);
Hs = 20*log10(1/Xspeak);
fprintf('Original peak = %f\n',Xpeak);
fprintf('Scaled peak = %f\n',Xspeak);
fprintf('Original headroom = %.4f dB\n',H);
fprintf('Scaled headroom = %.4f dB\n',Hs);
figure;
plot(y);
hold on;
Figures
Figure 1
Effect of Signal Scaling
Unscaled
Scaled
Amplitude
Sample
Code Issues
Filter
No issues to display


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EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
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mw 1.m
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untitled.m uni
untitled.m 5g & 6g.m
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untitled3.m
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Ap 1 1×1 double
Ast 40 1×1 double
b 1×13 double 1×13 double
Fp 100 1×1 double
Fs 1000 1×1 double
Fst 150 1×1 double
N 12 1×1 double
overflow... 0 1×1 double
overflow... 4 1×1 double
Editor
untitled26.m
untitled27.m
untitled28.m
untitled29.m
/MATLAB Drive/untitled29.m
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y = filter(b,a,x);
ys = filter(b,a,xs);
% Q1.7
S = 128;
yq = round(y*S)/S;
ysq = round(ys*S)/S;
overflow_unscaled = sum(abs(yq)>1);
overflow_scaled = sum(abs(ysq)>1);
fprintf('Unscaled peak = %f\n',max(abs(yq)));
fprintf('Scaled peak = %f\n',max(abs(ysq)));
fprintf('Unscaled overflow samples = %d\n',overflow_unscaled);
fprintf('Scaled overflow samples = %d\n',overflow_scaled);
figure;
plot(yq);
Figures
Figure 1
Numerical Response Before and After Scaling
Unscaled
Scaled
Amplitude
Sample
Code Issues
Filter
No issues to display
Editor: 80%
UTF-8
LF
Script
Ln 1 Col 1
26°C
Light rain
Search
ENG
IN
11:34 PM
16-10-2026